Saeid Chahardoli

Baton Rouge, LA, USA · +1 (225) 409-9216 · schaha1@lsu.edu
GitHub · LinkedIn · saeidchahardoli.com · Google Scholar

Available Summer 2027 · Eligible for CPT/OPT work authorization

SUMMARY

AI and robotics researcher with an M.S. in Computer Science and an engineering background, working on deep reinforcement learning, physics-informed and sequence models, and autonomous mobile robots for physical systems. Shipped a production forecasting and retrieval-augmented generation platform; built transfer learning with a Transformer Adapter.
35 indexed publications, 109 citations, h-index 5.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C/C++, MATLAB

ML & Deep Learning: PyTorch, scikit-learn, XGBoost; Transformers and RNN/LSTM for time series; reinforcement learning (PPO); Bayesian optimization; tabular foundation models (TabPFN v2, TabICL); physics-regularized networks (KAN); proper orthogonal decomposition and surrogate modeling

LLMs & Agents: Retrieval-augmented generation, agentic workflows, vector retrieval, prompt engineering

Robotics: ROS 2, SLAM, Nav2, sensor fusion and integration (CO_2, T/RH, anemometer, LiDAR, IR camera)

Data & Infrastructure: Pandas, NumPy, Elasticsearch, SQL Server, Azure, GitHub Actions, Docker, Git, Linux, MQTT, Jupyter

Simulation & Domain: ANSYS Fluent, EnergyPlus, CFD post-processing, building energy and HVAC modeling

EXPERIENCE

Data Scientist / AI Intern | Hart Howerton, San Francisco, CA *June – August 2026*

- Built and deployed a live BIM model-health platform ingesting Revit authoring telemetry from BIMbeats (Elasticsearch), Guardian (SQL Server), Revit journal files, Autodesk Construction Cloud, and Airtable.
- Automated the ETL on GitHub Actions on a three-hour schedule with pyRevit document-sync hooks, replacing a manual model-audit process, and served the results as a web application on Azure.
- Benchmarked seven walk-forward forecasting methods, including tabular foundation models (TabICL, TabPFN v2), to predict which projects would breach model-health thresholds; the best model reached 0.245 PR-AUC against a 0.054 baseline, a 4.5× lift, validated across 871 project-months.
- Built a retrieval-augmented generation layer delivering weekly personalized Revit training to modelers, plus a digest flagging at-risk projects to the design-technology team.

Graduate Research Assistant | Louisiana State University *2023 – Present*

Deep Reinforcement Learning for Real-Time Simulation Calibration

- Trained a PPO agent to tune simulation boundary conditions against live sensor streams from an instrumented chamber, cutting manual calibration from hours to minutes per case and reducing velocity-field error versus a hand-tuned baseline (ASHRAE Transactions, 2026).
- Designed an RL ventilation-control policy over a joint thermal-comfort and energy reward, reducing energy use relative to rule-based control at equal occupant comfort (Computing in Civil Engineering, 2025).

Active Learning and Bayesian Optimization for Sensor Placement & Data Collection

- Built an uncertainty-quantification loop that actively selects next-best measurement locations from a mobile sensing platform, replacing fixed-grid sampling and reducing the number of measurements needed per zone (E3S Web of Conferences, 2026).
- Built a custom ROS 2 mobile robot logging CO_2, temperature, RH, and airflow with hardware time synchronization, SLAM mapping, and Nav2 waypoint navigation, producing spatially registered datasets across multiple indoor zones.

Physics-Informed and Sequence Models for Spatiotemporal Prediction

- Trained Transformer and BiLSTM models for multi-step forecasting of indoor environmental time series, fine-tuned using Transformer Adaptor on local chamber experiments and validated against measured hold-out data.
- Developed physics-regularized Kolmogorov–Arnold and reduced-order surrogates predicting full concentration fields in real time, cutting per-configuration evaluation from 55 to 15 minutes (ASHRAE Transactions, 2026; E3S Web of Conferences, 2026).

Combinatorial Bayesian Optimization and Multimodal Comfort Modeling

- Applied combinatorial Bayesian optimization over a mesh-sensing network to place chamber air purifiers, finding a near-optimal configuration in a small fraction of the evaluations required for exhaustive search (E3S Web of Conferences, 2026).
- Trained models predicting individual thermal comfort from IR imaging and wearable sensor streams, trained and evaluated on measured participant data (REHVA HVAC World Congress, 2025).

Lecturer | Various universities, Iran *2018 – Present*

- Delivered 30+ technical sessions across multiple institutions.

EDUCATION

Ph.D. Candidate, Construction Management | Louisiana State University *2023 – Present*

Research area: Deep Reinforcement Learning and DNN integration with EBM to predict indoor physical fields in real time.
Coursework: DL, ML, RL, Bayesian optimization, LLMs, Agentic AI.

M.S., Computer Science | Louisiana State University *2024 – 2026*

Thesis: Transfer Learning with a Transformer Adapter for Personalized Thermal Comfort Prediction Using Sensors.
Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Agentic AI, Large Language Models (RAG).

M.S., Architectural Engineering | Iran University of Science and Technology, Tehran, Iran *2017 – 2020*

B.S., Architectural Engineering | Malayer University, Hamadan, Iran *2013 – 2017*

SELECTED PUBLICATIONS

9 of 35 indexed publications; 117 citations, h-index 5. Full list: Google Scholar

Machine Learning, Reinforcement Learning & Autonomous Systems

- **Chahardoli, S.**, Lesan, M., Sarker, A., Navazani, M., & Bhattacharya, A. (2026). Real-time CFD model calibration of an environmental chamber using deep reinforcement learning. *ASHRAE Transactions*, 132, 1233–1240.
- **Chahardoli, S.**, Lesan, M., & Bhattacharya, A. (2025). A reinforcement learning-based ventilation strategy for occupant-centric thermal comfort and energy efficiency. *Computing in Civil Engineering 2025*.
- Sarker, A., Lesan, M., **Chahardoli, S.**, & Bhattacharya, A. (2026). Real-time indoor CO_2 distribution prediction using CFD and a physics-regularized Kolmogorov–Arnold network. *E3S Web of Conferences*, 716, 01039.
- **Chahardoli, S.**, Sarker, A., Lesan, M., Xiao, Y., Johannsen, A. Z., & Bhattacharya, A. (2026). SIMOC-inspired mesh sensing and combinatorial Bayesian optimization of chamber air-purifier placement. *E3S Web of Conferences*, 716, 01047.
- Xiao, Y., **Chahardoli, S.**, Lesan, M., Sarker, A., Johannsen, A. Z., & Bhattacharya, A. (2026). Optimizing ventilation inlet and outlet configurations using a UQ-informed robotic sensing platform. *E3S Web of Conferences*, 716, 01049.
- **Chahardoli, S.**, Zhu, Y., & Bhattacharya, A. (2025). Data-driven prediction of thermal comfort using IR imaging and wearable sensor data. *REHVA HVAC World Congress*, 530–540. **[1 citation]**
- **Chahardoli, S.**, Mousavi, E., & Bhattacharya, A. Indoor airflow patterns for random occupant movement prediction using an event-based approach. *SSRN preprint 6847975*.

Selected Journal Articles

- **Chahardoli, S.**, Sedghikhanshir, A., Chen, Y., Zhu, Y., & Bhattacharya, A. (2026). Experimental and numerical analysis of evaluating the cooling potential of a green wall in a controlled chamber. *Building and Environment*, art. 115119. **[Q1]**
- **Chahardoli, S.**, Khakzand, M., Faizi, M., & Siavashi, M. (2022). Numerical analysis of the effect of roof types and porch on particle dispersion and deposition around a low-rise building. *Journal of Building Engineering*, 53, 104533. **[Q1 · 28 citations]**

HONORS & AWARDS

- Scholarship award, IAQVEC Conference 2026, Los Angeles.
- 2nd place, poster presentation, Department of Construction Management, Louisiana State University, 2025.
- Full tuition waiver, Iran University of Science and Technology; ranked top 0.25% nationally in the Iranian Graduate University Entrance Exam.